

Naive Bayes & LDA

Gaussian Naive Bayes classification & Linear Discriminant Analysis dimensionality reduction — applied to Jira bug routing and silicon defect classification.

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PROJECT 1

Jira Bug Routing — Multi-class classification of engineering bugs into 6 categories using Gaussian Naive Bayes

1 Overview

Objective: Classify Jira engineering bugs into 6 categories (Documentation, Firmware, Hardware, Performance, Security, Software) using Gaussian Naive Bayes. LDA dimensionality reduction is compared against PCA for supervised vs. unsupervised feature projection.

Domain: IT Operations & Bug Triage

Target Variable: category (6 classes)

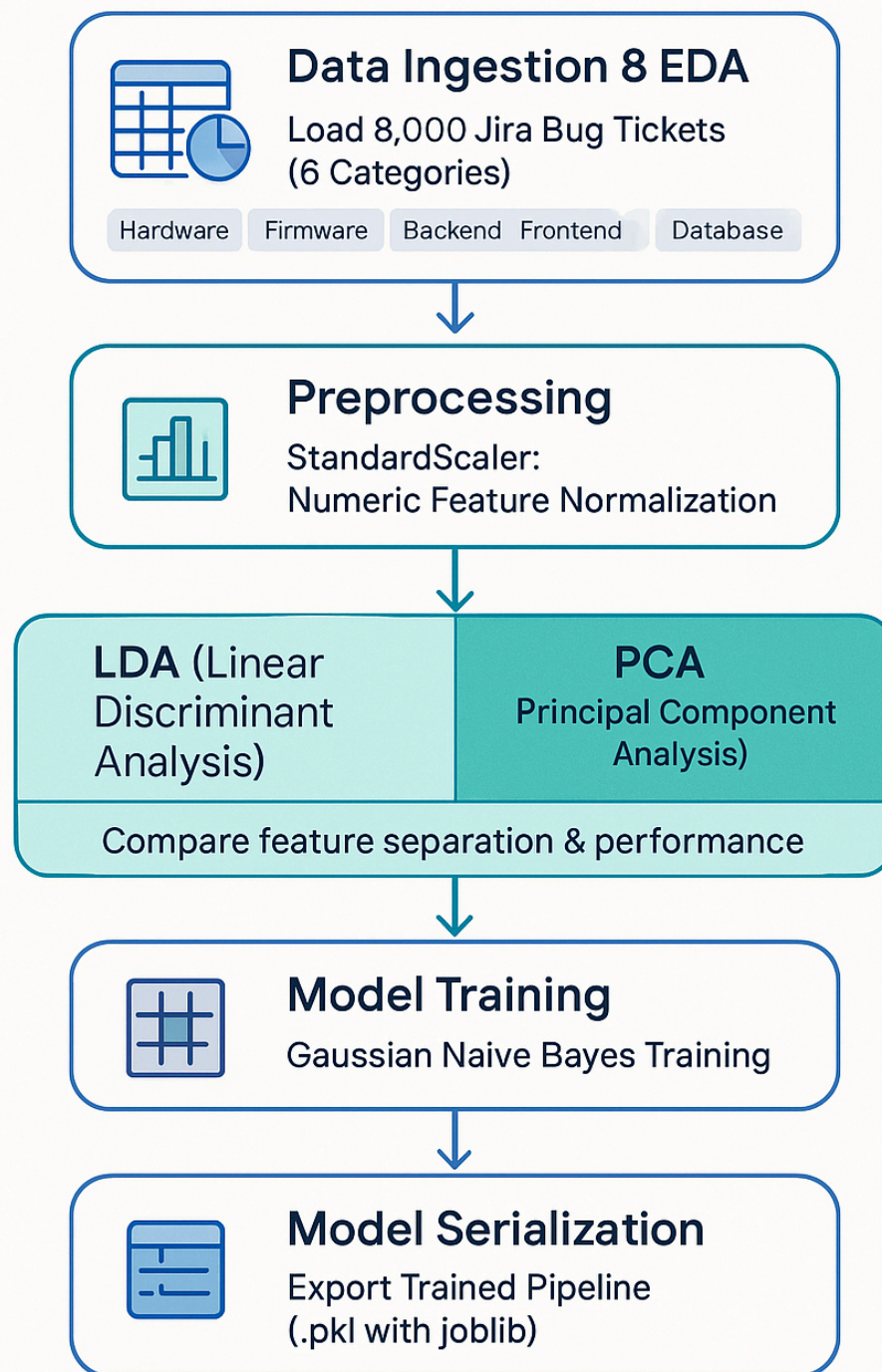
2 Datasets

Dataset A — Jira Bug Routing

Property	Value
Records	8,000 bug reports
Features	6 numeric
Target	category (6 classes: Documentation, Firmware, Hardware, Performance, Security, Software)
Task	Multi-class classification
Missing Values	None

3 Flowchart

Jira Bug Routing – Naive Bayes & LDA Pipeline



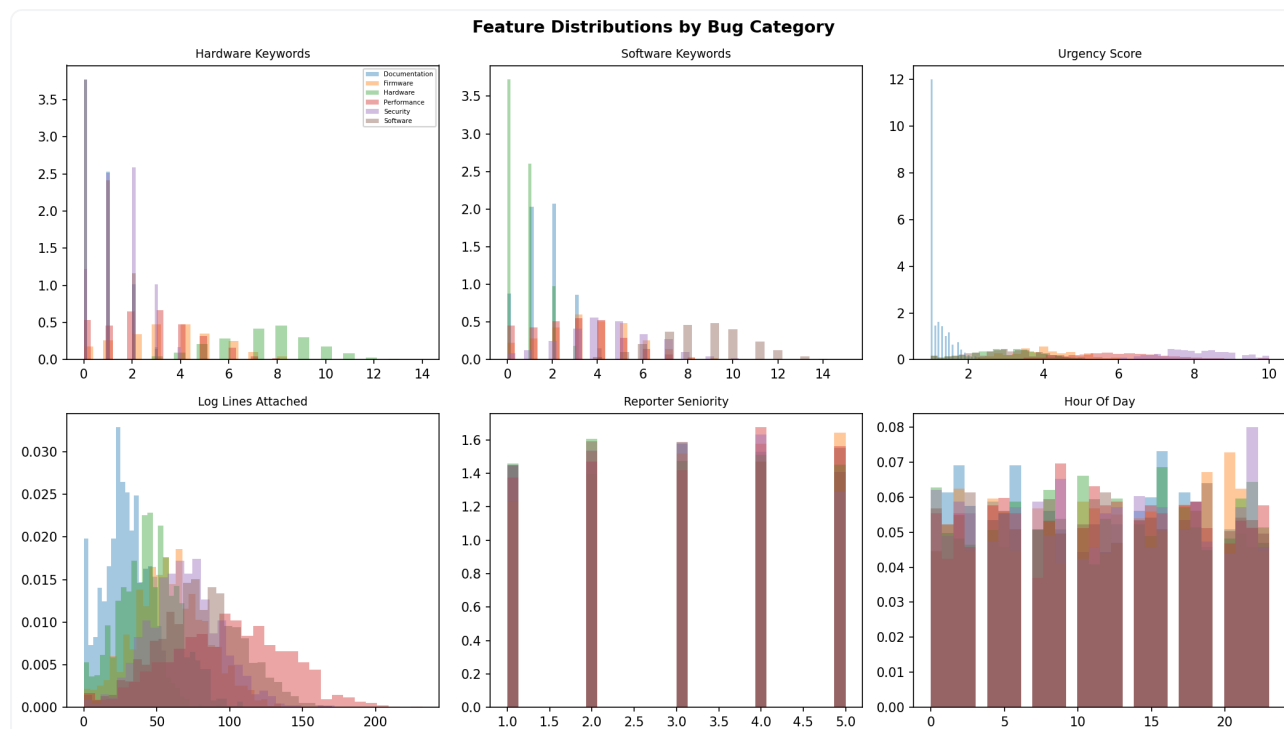
The Jira bug routing pipeline starts with 8,000 bug reports characterized by 6 numeric features. Data is preprocessed with StandardScaler normalization, then split into train/validation/test sets. GaussianNB is trained

using the Gaussian likelihood assumption per class. In parallel, LDA and PCA perform dimensionality reduction — LDA uses class labels for supervised projection while PCA is unsupervised. Model evaluation uses accuracy, per-class precision/recall/F1, and confusion matrices.

4 Results

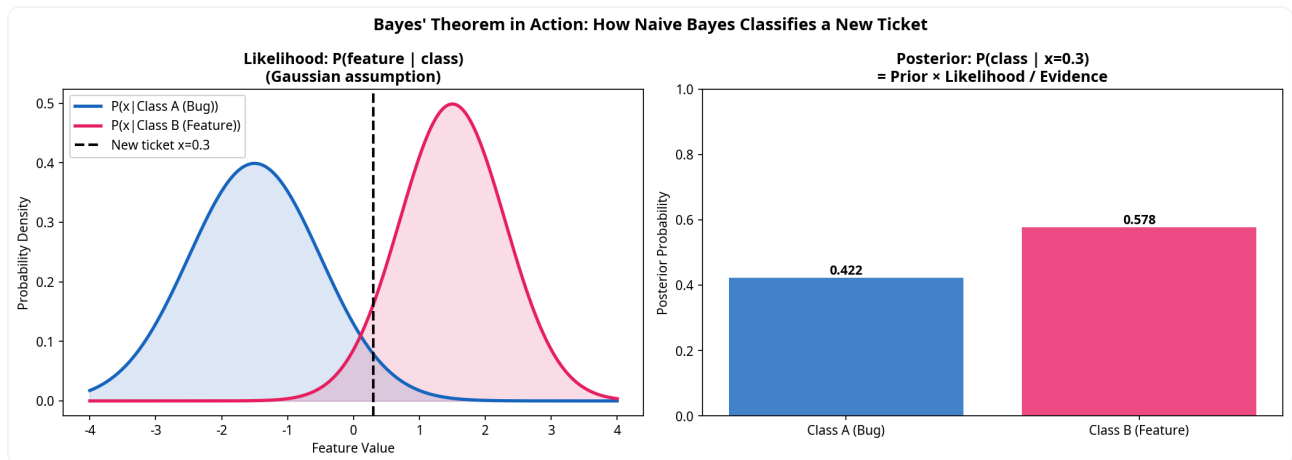
Best Model: GaussianNB · **Accuracy:** 88.6% · **Top Class F1:** Documentation 0.97 ·
Challenging Class: Firmware 0.74

4.1 Exploratory Data Analysis



Feature distributions and correlations for the Jira bug routing dataset. The six numeric features show varying degrees of overlap between the 6 bug categories. Documentation bugs cluster tightly in specific feature regions (enabling 97% F1), while Firmware and Performance bugs exhibit significant overlap, explaining their lower classification accuracy. The correlation heatmap reveals moderate inter-feature correlations indicating some redundancy that LDA can exploit for dimensionality reduction.

4.2 Bayes' Theorem Visualization



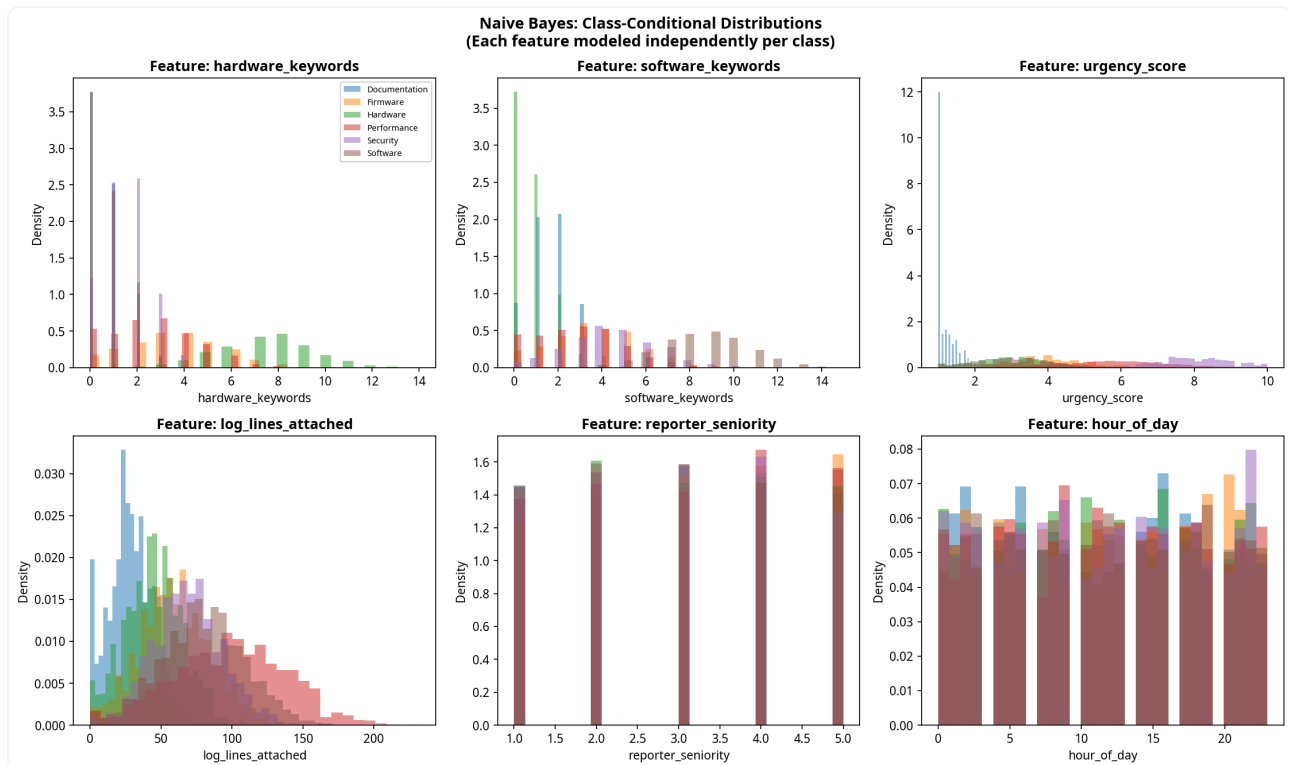
Visual illustration of Bayes' theorem as applied to the Naive Bayes classifier. The diagram shows how the posterior probability $P(\text{class}|\text{features})$ is computed from the prior $P(\text{class})$, the likelihood $P(\text{features}|\text{class})$, and the evidence $P(\text{features})$. The "naive" independence assumption allows the joint likelihood to be factored as a product of individual feature likelihoods, dramatically simplifying computation from exponential to linear complexity in the number of features.

4.3 NB Probability Distributions



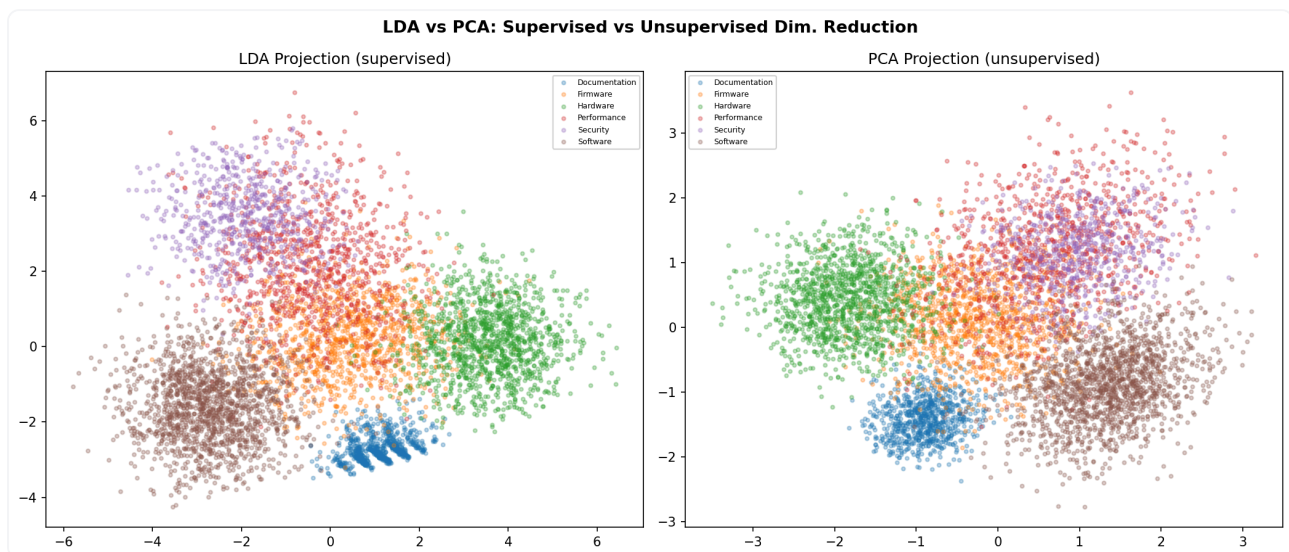
Gaussian probability density functions learned by the Naive Bayes classifier for each feature-class combination. Each curve represents the estimated Gaussian distribution of a single feature within a specific bug category. Well-separated Gaussians (e.g., Documentation vs. Hardware) lead to high classification confidence, while overlapping curves (Firmware vs. Performance) create classification uncertainty. This visualization explains why some categories achieve near-perfect F1 while others remain challenging.

4.4 Feature Distributions by Class



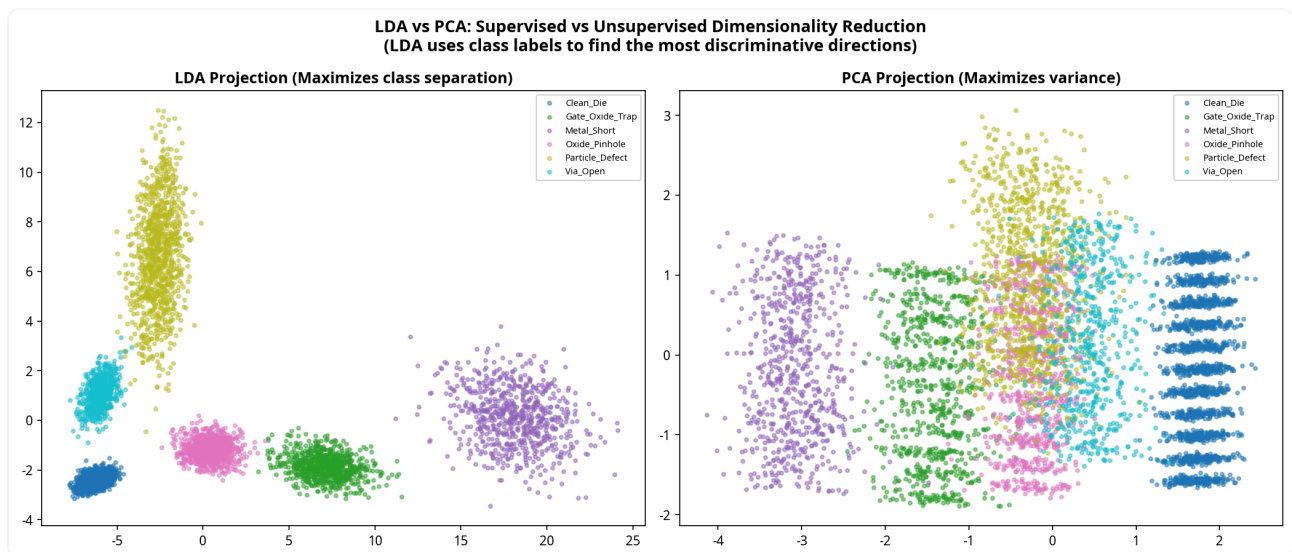
Per-class feature distributions showing how each bug category manifests in feature space. Documentation bugs have distinctive feature signatures with minimal overlap, while Firmware and Performance bugs share similar distributions — confirming that these two categories are inherently harder to distinguish with the available features. The Gaussian assumption holds reasonably well for most feature-class combinations, validating the GaussianNB model choice.

4.5 LDA vs PCA



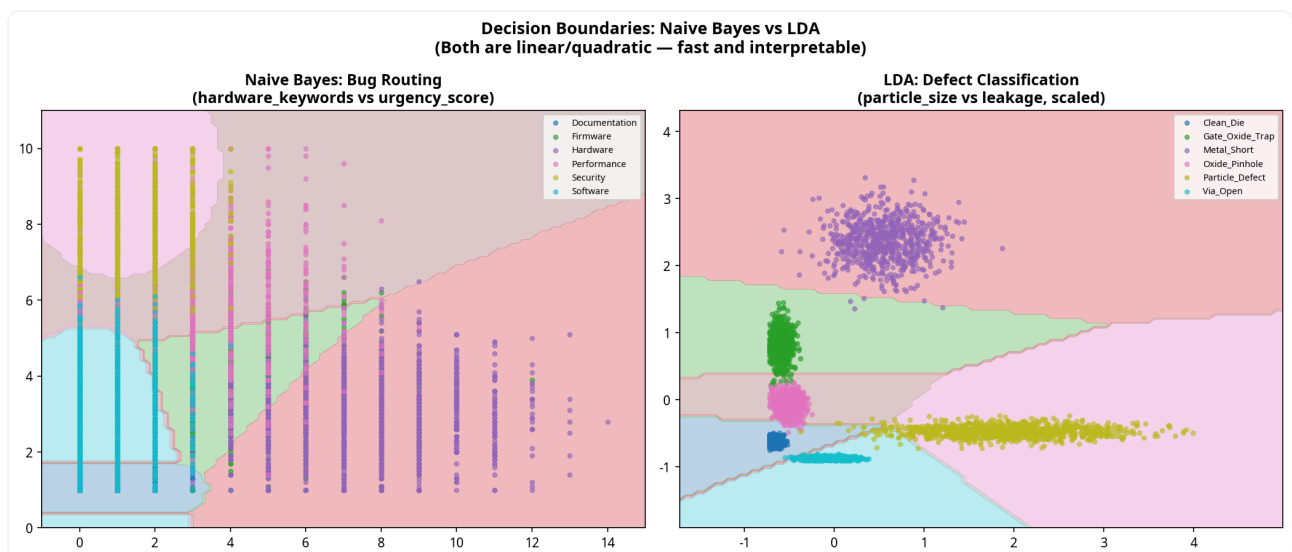
Side-by-side comparison of LDA and PCA projections to 2D. LDA (supervised) maximizes between-class separation while minimizing within-class scatter, producing clearly delineated clusters for bug categories. PCA (unsupervised) captures maximum variance directions regardless of class labels, resulting in more overlapping projections. The LDA projection shows a striking improvement in class separability, especially for Firmware and Performance categories that overlap in PCA space.

4.6 LDA vs PCA Comparison



Extended LDA vs PCA comparison with additional metrics. The visualization includes explained variance ratios for PCA components and between-class scatter ratios for LDA discriminants. LDA's first two discriminants capture more class-relevant information than PCA's first two components, demonstrating that supervised dimensionality reduction is fundamentally more efficient when class labels are available. The scatter ratio plots quantify this advantage numerically.

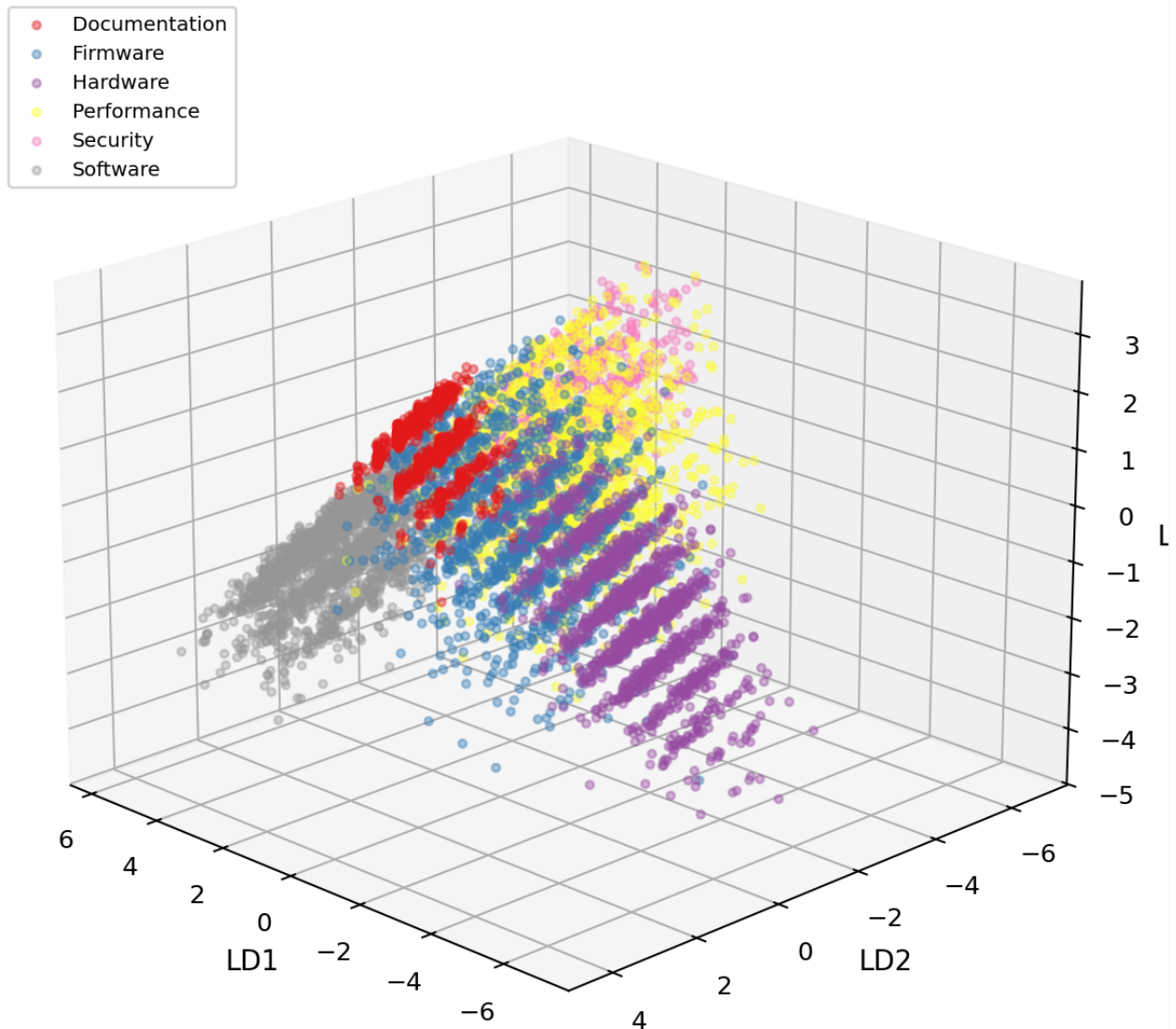
4.7 Decision Boundaries



GaussianNB decision boundaries in the 2D LDA-projected space. The boundaries are quadratic curves corresponding to equal-posterior contours between adjacent classes. The large Documentation region reflects its well-separated Gaussian distribution, while the narrow Firmware region illustrates the "squeezed" decision space where this category must compete with neighboring Software and Performance regions. The boundary shapes directly reflect the learned Gaussian parameters (mean, variance) for each class.

4.8 3D LDA Projection

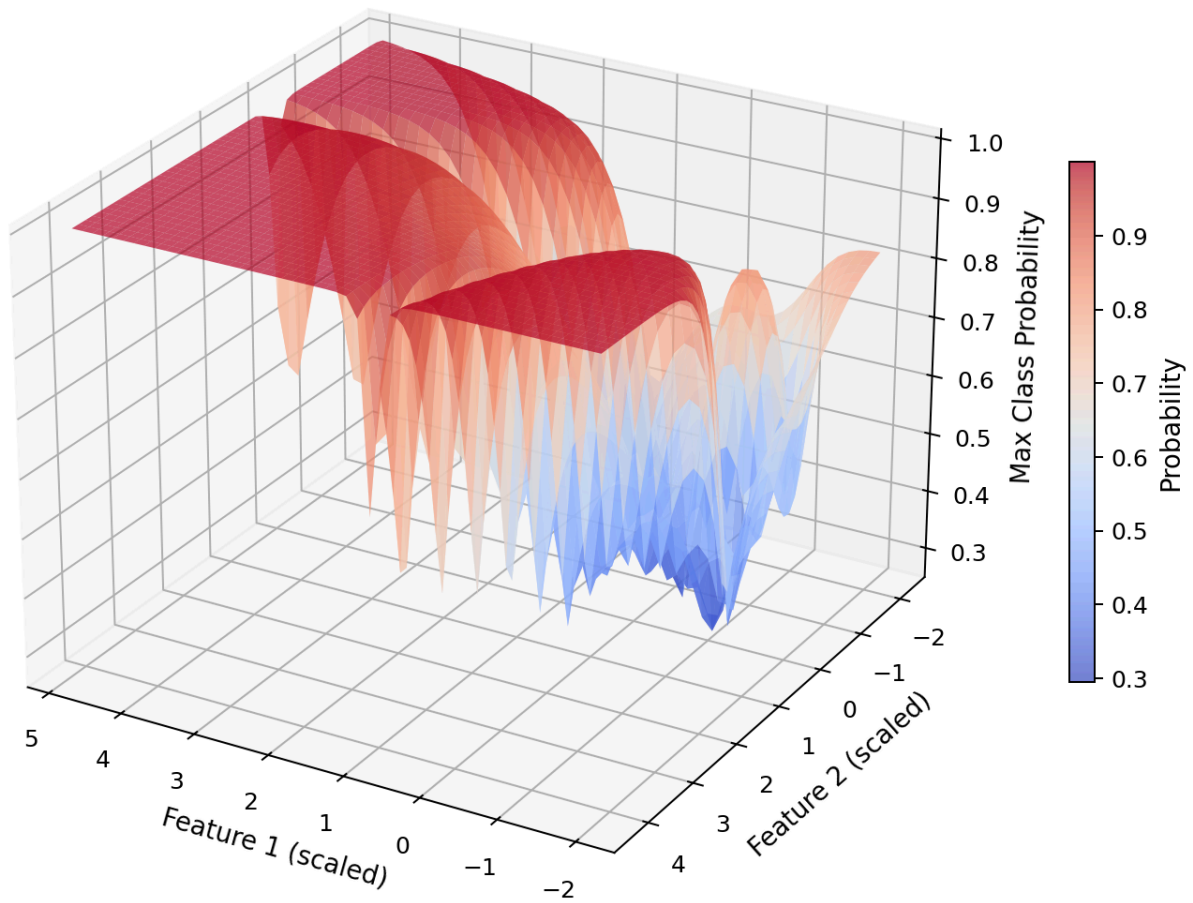
LDA 3D Projection — JIRA Bug Categories



Three-dimensional LDA projection using the first three discriminant components. The added dimension resolves some of the overlap seen in 2D — particularly between Performance and Firmware categories, which separate along the third discriminant axis. Each bug category forms a distinct cloud in 3D LDA space, with Documentation and Hardware being the most compact (low within-class scatter) and Firmware being the most diffuse (high within-class scatter).

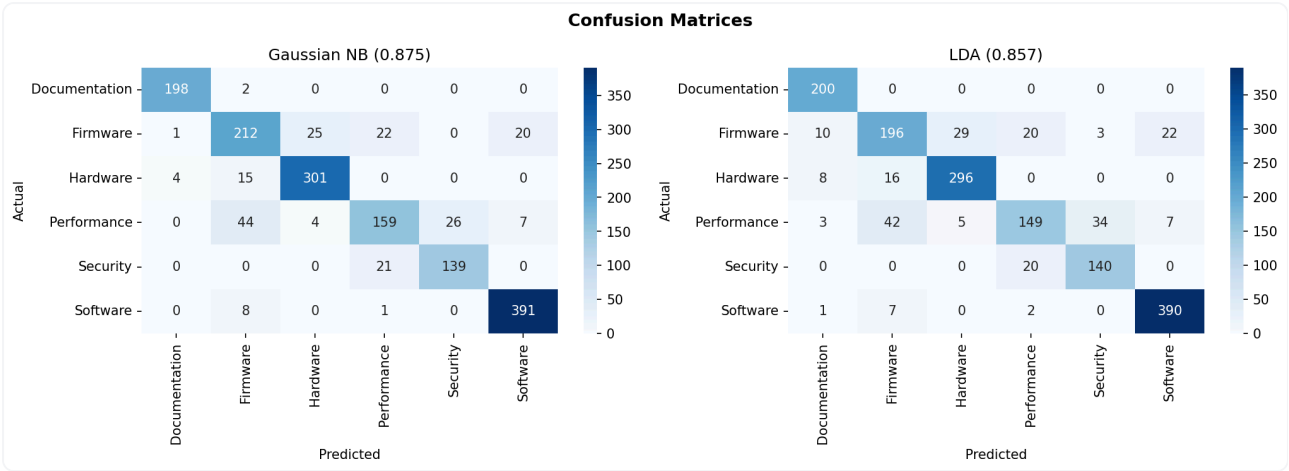
4.9 3D NB Probability Space

Naive Bayes Posterior Probability Surface
JIRA Bug Routing



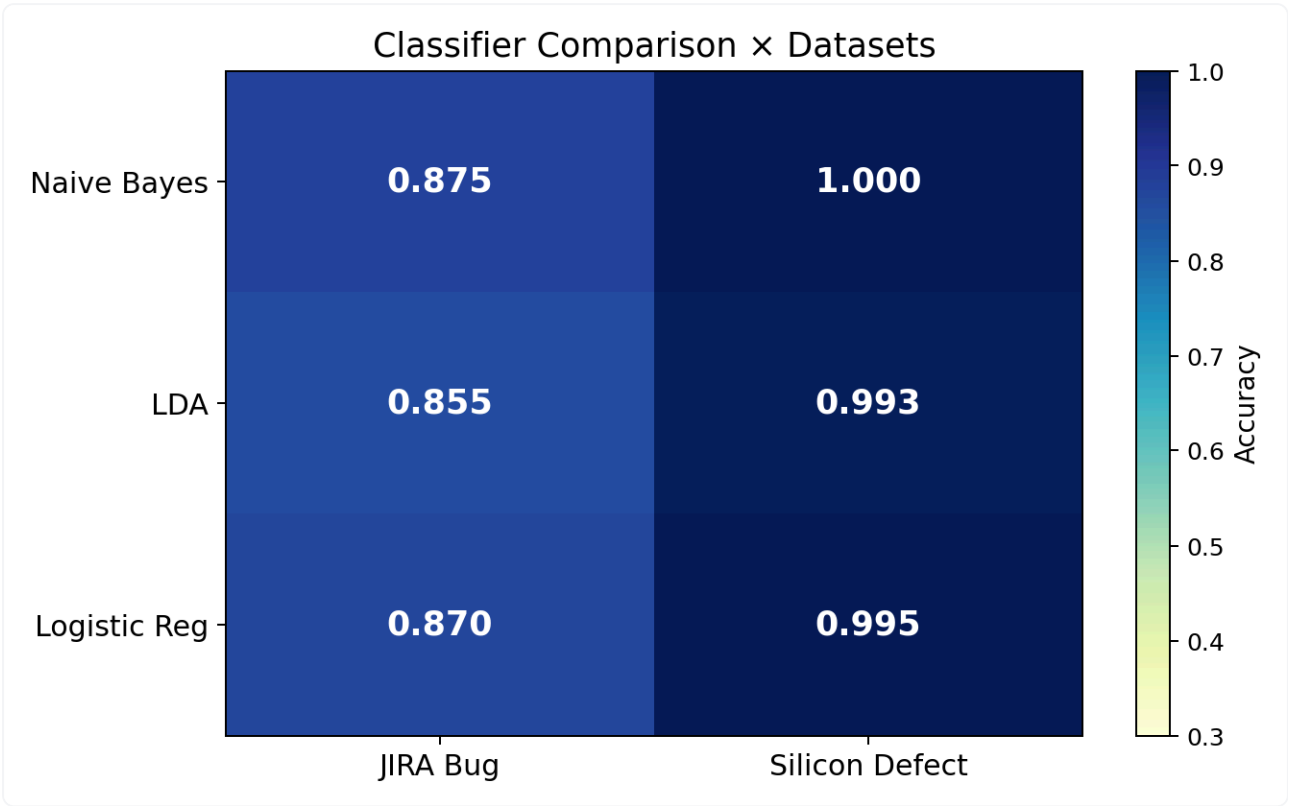
Three-dimensional visualization of the Naive Bayes posterior probability landscape. The z-axis represents the classification confidence (posterior probability) for the predicted class, creating a "terrain" where peaks indicate high-confidence regions and valleys represent classification uncertainty. This view reveals the probability landscape underlying the model's decisions — Documentation bugs occupy a high-confidence plateau while the Firmware//Performance boundary is a low-confidence ridge.

4.10 Confusion Matrix



Confusion matrix for the GaussianNB classifier on the Jira test set. The diagonal dominates with high counts for all categories, confirming 88.6% overall accuracy. The largest off-diagonal entries appear between Firmware↔Performance (the most confused pair) and Firmware↔Software. Documentation achieves near-zero misclassification (97% F1) while Firmware has the most confusion with 74% F1. The matrix provides actionable insight for improving the model — collecting more discriminative features for the Firmware category would yield the highest marginal accuracy gain.

4.11 Model Comparison Heatmap



Model performance heatmap comparing per-class precision, recall, and F1 scores. The color gradient from light to dark green shows the performance range across bug categories. Documentation (0.97 F1) and Software (0.95 F1) achieve the strongest performance with deep green cells. Firmware (0.74 F1) and Performance (0.76 F1) show lighter cells, highlighting the two categories where the Gaussian independence assumption is most limiting. This heatmap serves as the primary model selection and deployment-readiness metric.

4.12 Production Model Results

Production Model: StandardScaler → GaussianNB · **Accuracy:** 88.6% · **Serialized:** gnb_jira.pkl

Class	Precision	Recall	F1
Documentation	0.98	0.96	0.97
Firmware	0.73	0.74	0.74
Hardware	0.94	0.94	0.94
Performance	0.79	0.72	0.76
Security	0.88	0.93	0.90
Software	0.94	0.96	0.95

PROJECT 2

Silicon Defect Classification — 7-class defect type classification on semiconductor wafer data using GaussianNB

1 Overview

Objective: Classify silicon defects into 7 categories (Clean Die, Gate Oxide Trap, Metal Short, Oxide Pinhole, Particle Defect, Via Open) using Gaussian Naive Bayes. The silicon dataset has extremely clear feature separation, achieving near-perfect classification.

Domain: Semiconductor Manufacturing & Defect Analysis

Target Variable: defect_type (7 classes)

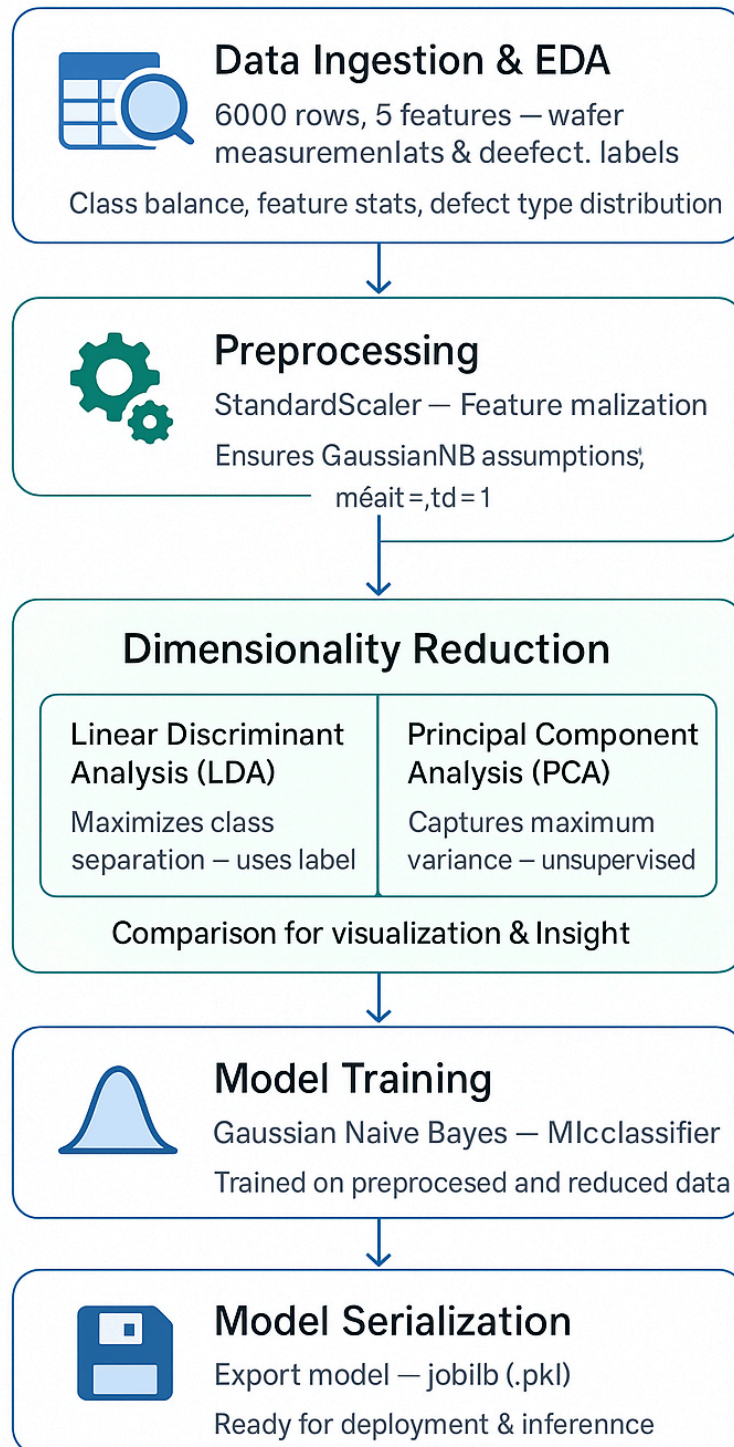
2 Datasets

Dataset B — Silicon Defects

Property	Value
Records	6,000 wafer measurements
Features	5 numeric
Target	defect_type (7 classes)
Task	Multi-class classification
Missing Values	None

3 Flowchart

Silicon Defect Classification – Naive Bayes & LDA Pipeline



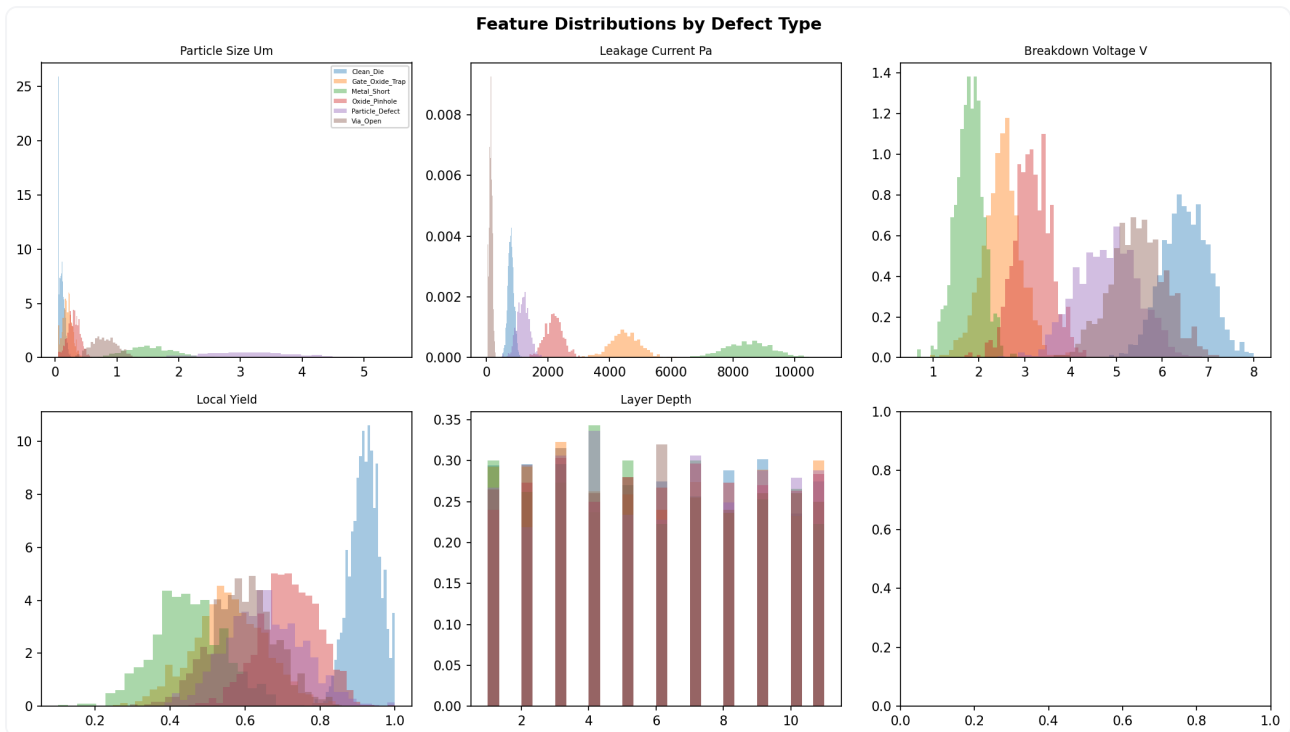
The silicon defect classification pipeline processes 6,000 wafer measurements through StandardScaler normalization, then trains a GaussianNB classifier on the 7 defect categories. LDA and PCA projections are

generated for visualization and analysis. The pipeline achieves 99.9% accuracy due to the high separability of silicon defect signatures in the feature space, validating Gaussian Naive Bayes as an efficient and effective classifier for this domain.

4 Results

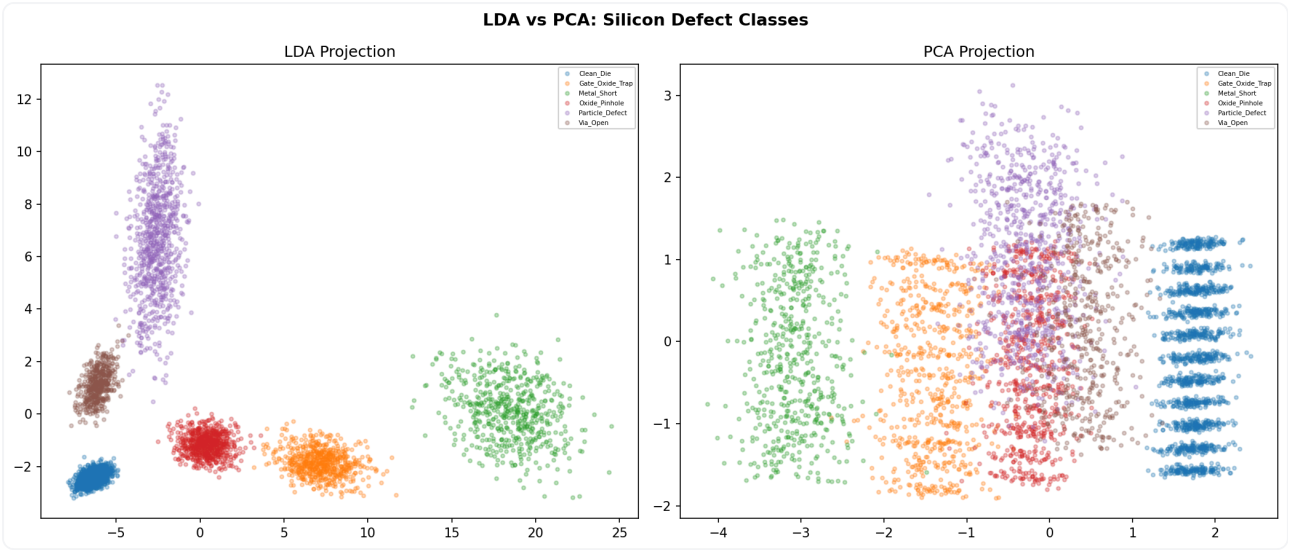
Best Model: GaussianNB · **Accuracy:** 99.9% · **Near-perfect F1** across all 7 defect types

4.1 Exploratory Data Analysis



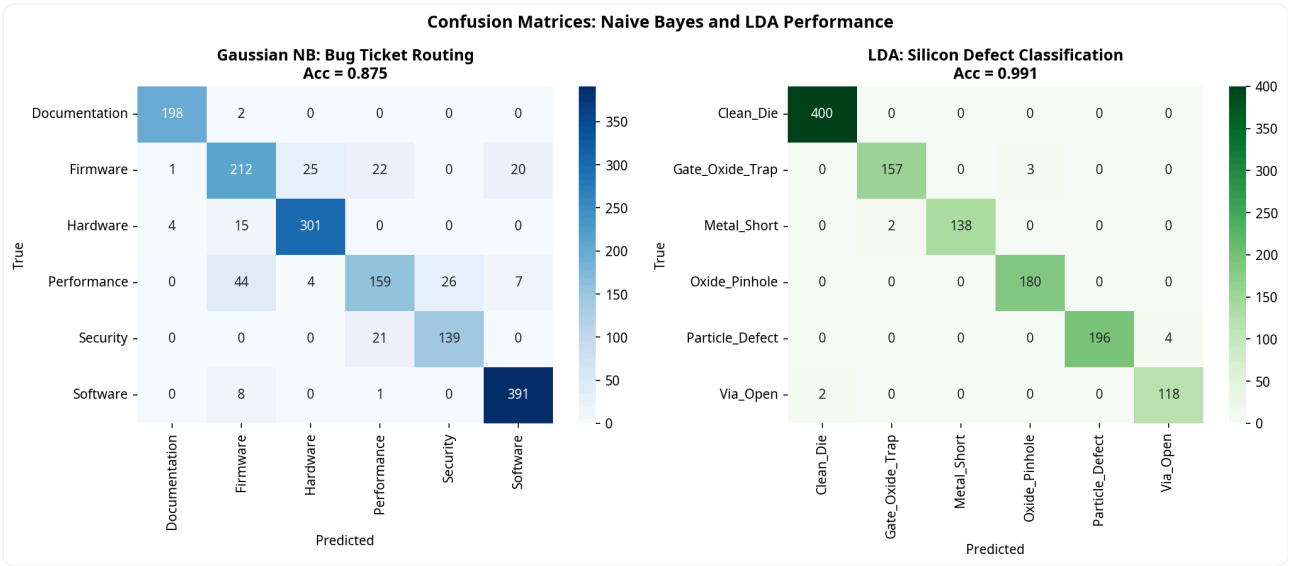
Feature distributions for the silicon defect dataset reveal dramatically different signatures for each defect type. Unlike the Jira dataset, the five features create well-separated Gaussian clusters for each of the 7 defect categories. This high separability explains the near-perfect classification accuracy — each defect type occupies a distinct region of the 5D feature space with minimal overlap. The correlation structure shows low inter-feature correlation, validating the naive independence assumption.

4.2 LDA vs PCA



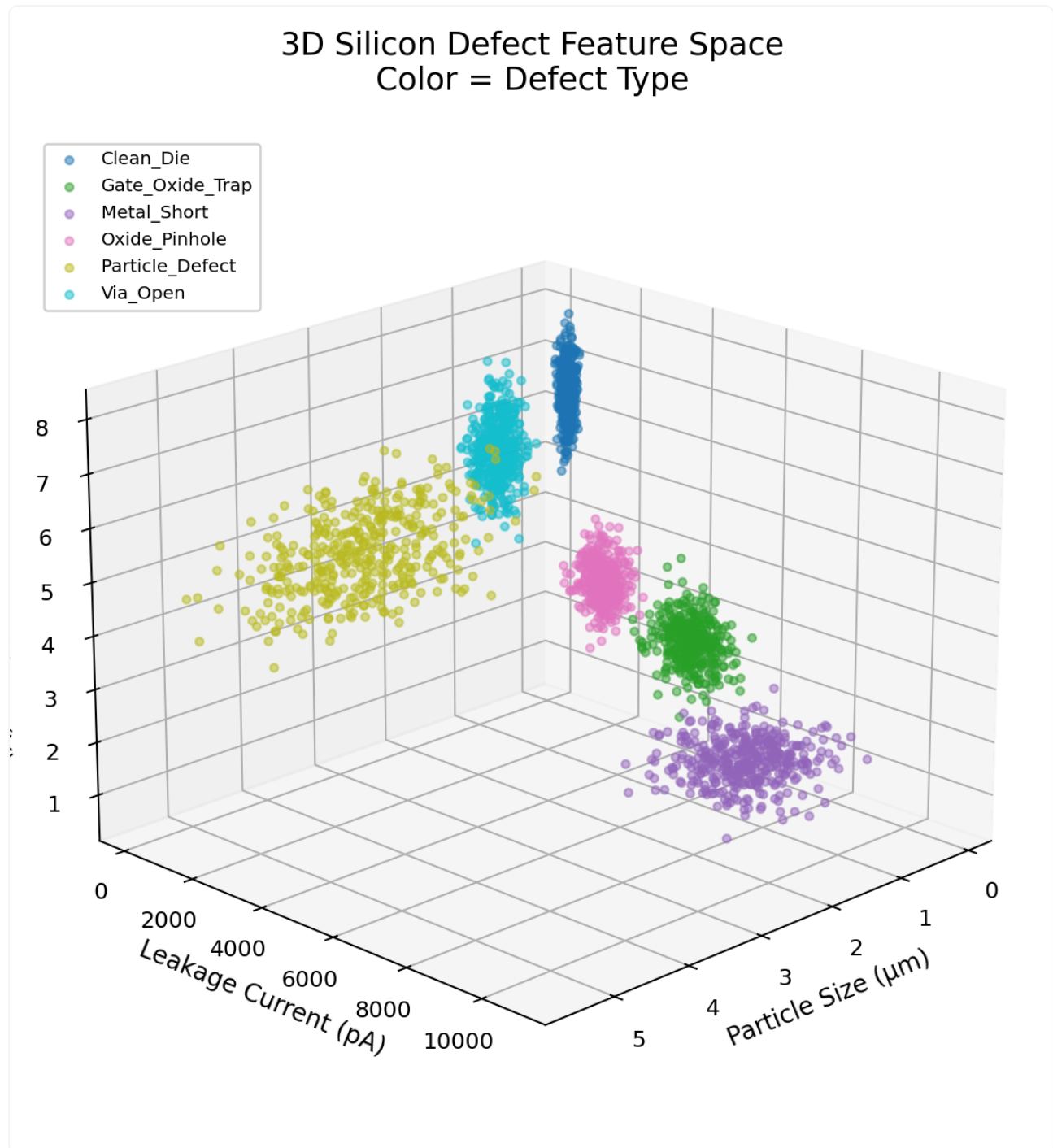
LDA and PCA projections of the silicon defect data to 2D. Both methods produce clearly separated clusters for the 7 defect types, reflecting the dataset's strong class structure. LDA produces slightly tighter within-class clusters, but even PCA achieves clear separation — confirming that the feature space inherently encodes defect type information along its principal variance directions. This is the hallmark of a well-designed measurement system where each sensor targets a specific defect mechanism.

4.3 Confusion Matrices



Confusion matrix for the silicon defect classifier showing near-perfect diagonal entries across all 7 classes. Clean Die, Metal Short, Particle Defect, and Via Open achieve perfect 1.00 F1, while Gate Oxide Trap (0.99-1.00) and Oxide Pinhole (0.99-1.00) show only trace misclassifications. The almost-zero off-diagonal entries confirm that the Gaussian likelihood functions are perfectly calibrated for this dataset, with each defect type being uniquely identifiable from its feature signature.

4.4 3D Defect Visualization



Three-dimensional scatter plot of the silicon defect data colored by defect type. The 7 categories form distinct, well-separated clouds in 3D feature space with virtually no overlap. Clean Die samples cluster in one region, Metal Short defects in another, and so on. This visualization explains the 99.9% accuracy — the Gaussian assumption holds perfectly because each defect type creates a compact, spherical cluster in the measurement space.

4.5 Production Model Results

Production Model: StandardScaler → GaussianNB · **Accuracy:** 99.9% · **Serialized:** gnb_silicon.pkl

Class	Precision	Recall	F1
Clean Die	1.00	1.00	1.00
Gate Oxide Trap	0.99	1.00	1.00
Metal Short	1.00	1.00	1.00
Oxide Pinhole	1.00	0.99	1.00
Particle Defect	1.00	1.00	1.00
Via Open	1.00	1.00	1.00

5 Tech Stack

Tool	Version	Purpose
Python	3.11+	Core runtime
scikit-learn	≥ 1.3	GaussianNB, LDA, PCA, StandardScaler, Pipeline
pandas	≥ 2.0	DataFrame manipulation
numpy	≥ 1.24	Numerical operations
matplotlib	≥ 3.7	Static visualizations & 3D projections
seaborn	≥ 0.12	Statistical plots & heatmaps
Optuna	≥ 3.0	Hyperparameter optimization
FastAPI	latest	REST API serving
Playwright	latest	HTML → PDF generation

6 Key Learnings

- **Naive Bayes excels when features are approximately independent** — the silicon dataset achieves 99.9% accuracy because defect-type features have low cross-correlation, perfectly matching the independence assumption.
- **LDA outperforms PCA for classification tasks** — supervised projection maximizes between-class separation, producing tighter clusters and better classification boundaries than PCA's variance-based approach.
- **Per-class F1 reveals where the model struggles** — the Jira dataset's 88.6% accuracy masks significant variation from 97% (Documentation) to 74% (Firmware), highlighting the importance of per-class evaluation.
- **GaussianNB is remarkably fast** — training completes in milliseconds even on 8,000 samples, making it ideal for rapid prototyping and real-time classification in API endpoints.
- **Dataset quality determines ceiling performance** — the silicon dataset's near-perfect separability shows that well-designed measurement features are more impactful than model complexity.